

Drone-Remote Weapon Station

MIL-STD-882E System Safety

(Human-in-the-Loop)



1. System Safety Program

Plan (SSPP)

- Safety Manager: TBD
- Risk Acceptance Authority: TBD
- Safety Organization Structure: TBD
- Safety Reviews Schedule (SRR/PDR/CDR/TRR): TBD
- Lifecycle Safety Milestones: TBD

2. CONOPS SAFETY

- Normal Operation Description: TBD
- Arm/Disarm Logic: System Sequence of Operation
- Safe State Definition: TBD
- Behavior on Loss of Communication (weapon armed): Return Home
- Behavior on Power Loss: Return Home
- Behavior on Ammunition Jam: TBD
- Operator Authorization Logic: System Sequence of Operation
- Fail-safe Modes: Return Home

3. Risk Assessment Method

- Severity and probability categories per MIL-STD-882E with RAC classification.

4. Design Order of Precedence

- Elimination by design, risk reduction by design, safety devices, warnings/procedures.

3. Preliminary Hazard Analysis (PHA)

- Unintended fire
- Recoil Management instability
- Turret runaway
- Power loss
- Ammo cook-off
- Cyber/jamming
- Laser hazard
- GUI failure
- Sensor misID
- Feed failure
- Structural failure

4. Safety Assessment Report (SAR)

- Residual Risk Summary: TBD
- Risk Acceptance Sign-off: TBD

5. Detailed CONOPS Safety (Draft)

- Human-in-the-loop firing authorization required.
- Dual Channel-Channel Redundant Safety Control Architecture.
- Dual confirmation trigger.
- Automatic disarm on communication loss.
- Power-loss safe state.
- Jam detection inhibits firing.
- Mechanical stops prevent unsafe pointing.
- Fail of Recoil Management Mechanism.

6. Safety Data Package Statement

- This document constitutes a preliminary contractor system safety data package aligned with MIL STD 882E Sections 3 and 4.
- Residual risks subject to customer acceptance.

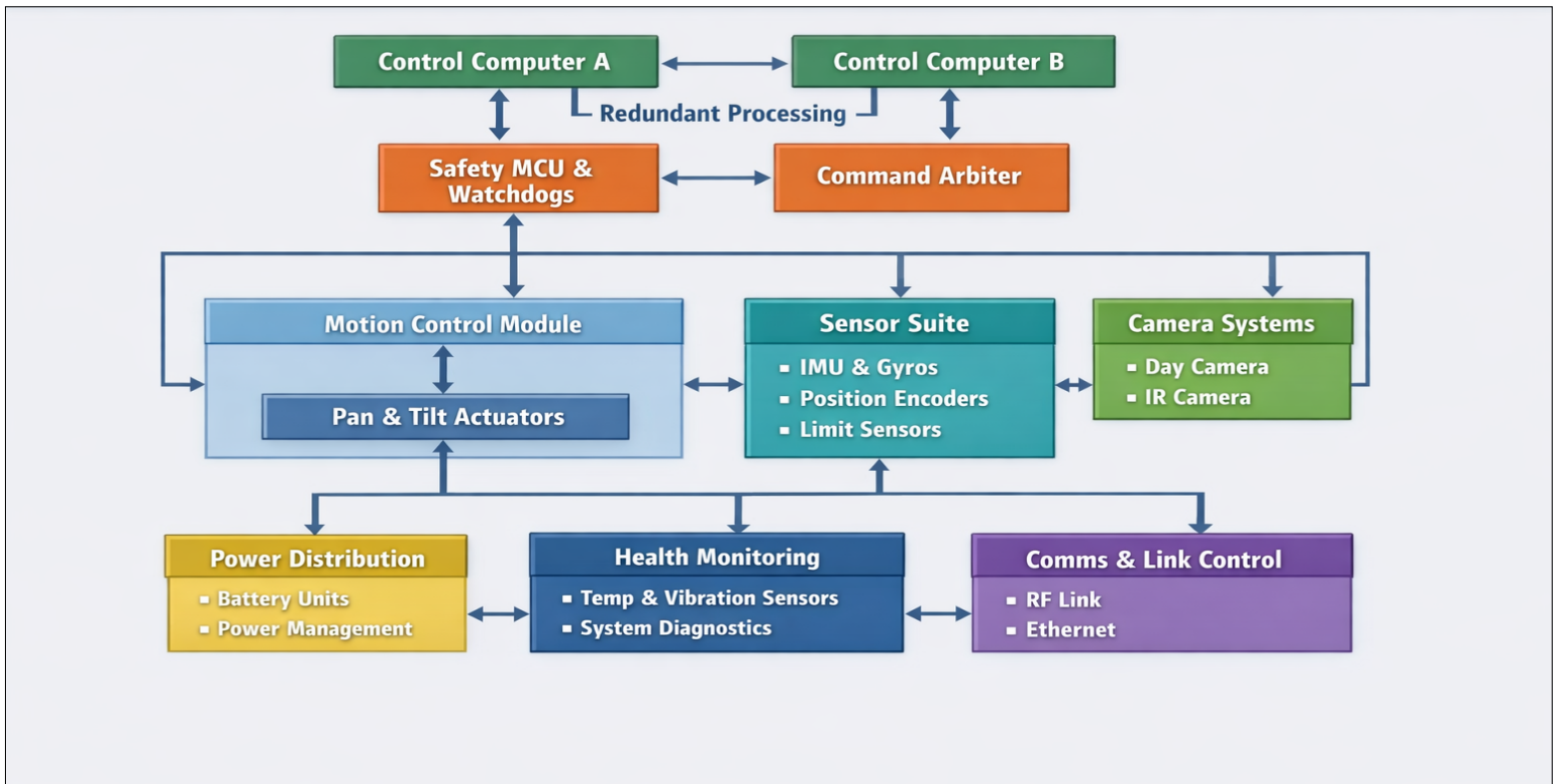
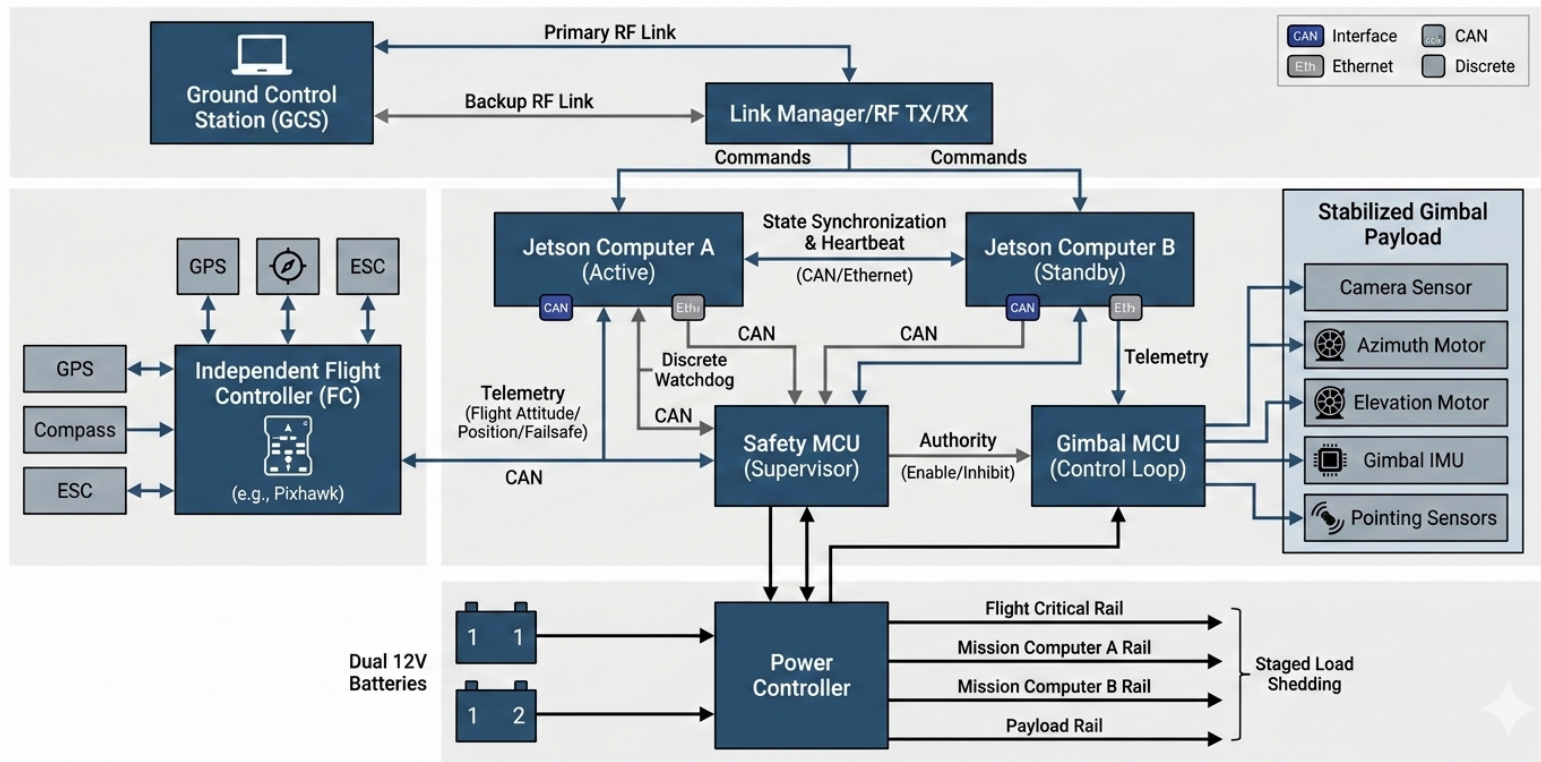
7. SHA Summary

- Subsystem interaction hazards between turret control, recoil mitigation, ammunition feed, DRONE flight control, power and communication interfaces.

8. General System Architecture

DUAL-REDUNDANT MISSION ARCHITECTURE - HIGH-LEVEL BLOCK DIAGRAM (Baseline)

Defined in the 'System Architecture Report (SAR)' Dual Redundant Payload' and file. 'general architecture.pptx'



9. Hazard Log – 40 Hazards, Preliminary RAC

The RAC values below are preliminary engineering estimates (Human-in-the-Loop RWS on DRONE).

N	Hazard / Unsafe Condition	Credible Cause(s)	Mishap / Effect	Initial RAC	Mitigation(s) (Design Devices Warnings/ Procedures)	Verification / Evidence
1	Unintended firing (SW logic fault)	Fire-control SW defect, mode-handling error	Injury/death; collateral damage	RAC 2 (Serious)	Independent ARM state machine; dual-confirm (operator + guarded ARM); two-channel FIRE command with plausibility checks; watchdog to safe; BIT on startup	SW unit/integration tests; fault-injection; HIL bench test; firing test with induced faults
2	Unintended firing (HW output stuck ON)	Relay/MOSFET short, driver failure	Injury/death; collateral damage	RAC 2 (Serious)	Series hardware inhibit (safety relay + mechanical sear/solenoid); monitored output feedback; crowbar/overcurrent protection; safe default on fault	Bench electrical fault tests (short/open); output feedback validation; firing inhibit test
3	Unintended firing (operator error)	Mode confusion; accidental trigger; UI ambiguity	Injury/death; collateral damage	RAC 1 (High)	Two-step FIRE (arm + confirm); guarded control; clear UI state; required positive ID & authorization; training & SOP; time-limited arm window	Human factors review; UI simulation; operator eval; procedural compliance drills
4	Unintended firing (spurious EM/ESD)	EMI coupling into trigger line; ESD	Injury/death; collateral damage	RAC 2 (Serious)	EMI/ESD filtering; shielded cables; debounce + command authentication; redundant inhibit; grounding/bonding	EMI/ESD tests (MIL-STD-461 as applicable); bench susceptibility tests
5	Firing while turret not in authorized sector	Sector limits mis-set; calibration drift	Friendly fire / collateral	RAC 2 (Serious)	Software geofencing/sector limits; hard stops; calibration check; inhibit FIRE outside sector; health monitoring	Range test; calibration verification; limit-switch tests; firing inhibit demo
6	Turret runaway (yaw)	Control loop instability; sensor fault; encoder slip	Unsafe pointing; structural impact	RAC 3 (Medium)	Rate/position limiters; watchdog; torque/current limiting; mechanical stops; redundant encoder plausibility	HIL stability tests; induced sensor fault; motion envelope tests
7	Turret runaway (pitch)	Control loop instability; IMU/encoder fault	Unsafe pointing; structural impact	RAC 3 (Medium)	As above + pitch hard-stop; gravity-compensation limits; safe park routine	Motion tests at extremes; fault injection; safe-park demo
8	Loss of comms while armed	RF link drop/jam; integration fault	Weapon remains armed without control	RAC 1 (High)	Heartbeat supervision; automatic DISARM on link loss; local timer to safe; operator re-auth required after reconnection	Comm-loss tests (RF off/jam); timing verification; flight test
9	Comm spoofing / unauthorized control	Weak auth; key compromise; spoofed packets	Unauthorized aiming/firing	RAC 2 (Serious)	Strong encryption + mutual auth; rolling keys; whitelist device IDs; command signing; anti-replay; secure boot	Pen-test; protocol tests; replay/spoof tests; code review
10	Latency / dropped packets causing wrong aim at FIRE moment	High latency; buffering; GUI lag	Miss / collateral	RAC 2 (Serious)	Time-stamped commands; aim-valid window; require stable track + operator confirm; inhibit if latency exceeds threshold; display latency indicator	Network emulation tests; end-to-end latency tests; firing drills
11	Stabilization failure during firing	Stabilizer control fault; IMU failure	DRONE instability; crash; collateral	RAC 2 (Serious)	Recoil mitigation; stabilization health monitoring; inhibit burst if stability out-of-bounds; reduce ROF/burst length; safe abort	Firing tests; IMU fault tests; stability threshold verification
12	Recoil mitigation failure (mechanical)	Damper leak/break; spring fracture	Loss of accuracy; DRONE upset	RAC 3 (Medium)	Mechanical margins; end-stops; condition monitoring; scheduled inspection; inhibit sustained fire on anomaly	Endurance firing test; inspection; vibration test
13	Mount structural failure	Fastener loosening; fatigue; overload landing	Weapon detaches; crash; injury	RAC 2 (Serious)	Structural analysis with safety factors; locking fasteners; secondary retention; periodic torque check; load limiters	Static load + fatigue test; vibration test; inspection
14	Weapon/munition overheating (cook-off)	High burst; blocked cooling	Cook-off; fire; injury	RAC 2 (Serious)	Thermal model; barrel temp sensor; firing duty-cycle limits; automatic cooldown inhibit; warnings/procedures	Thermal tests; firing schedule validation; sensor calibration
15	Ammunition box / feed chute jam	Feed chute damage; belt misalignment	Stoppage; unsafe clearing; damage	RAC 3 (Medium)	Jam detection; immediate trigger inhibit; jam-clear procedure with DISARM required; robust feed path; alignment guides	Jam tests; maintainability demo; induced misfeed trials
16	Runaway burst (trigger stuck)	Mechanical trigger linkage jam; solenoid	Extended firing; collateral	RAC 2 (Serious)	Burst limiter; cut-off timer; redundant inhibit; monitored firing	Burst-limit test; stuck-solenoid fault test; live-fire

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		stuck			feedback (shot counter) to stop if abnormal	demo
17	Failure to DISARM (arm latch stuck)	SW state latch; hardware interlock stuck	Weapon remains armed unexpectedly	RAC 3 (Medium)	Disarm in hardware + software; independent safety relay; disarm-on-fault; BIT; maintenance lockout	Fault-injection; bench tests; safety relay verification
18	Unexpected motion at power-up	No homing; encoder offset	Unsafe pointing at startup	RAC 2 (Serious)	Safe startup: power-up in SAFE; homing sequence; park pose; inhibit fire until homed; warnings	Power-cycle tests; homing validation; safe-start demo
19	Loss of power during armed state	Battery drop; connector intermittent	Unknown arm state; potential hazard	RAC 2 (Serious)	Fail-safe to DISARM on power loss; mechanical safe; state recorded; restart requires re-auth	Power interruption tests; connector wiggle tests
20	Battery thermal runaway / fire	Battery defect; overcurrent	Fire; loss of DRONE; collateral	RAC 2 (Serious)	BMS protection; fusing; thermal isolation; venting; temperature monitoring; charging controls	Abuse tests (as applicable); thermal monitoring validation; inspection
21	Laser pointer hazard (eye exposure)	Operator misuse; interlock failure	Eye injury	RAC 2 (Serious)	Laser class management; gated enable; interlock with aim zone; warning indicators; SOP; auto time-out	Laser safety test; interlock verification; operator training records
22	Rangefinder emits when not intended	UI bug; stuck enable	Eye hazard / emissions issue	RAC 3 (Medium)	Positive enable + indicator; time-out; separate control path; BIT	Functional tests; UI regression; emission enable tests
23	Sensor failure leads to incorrect aim indication	EO/IR freeze; stale frames	Operator mis-aim; collateral	RAC 2 (Serious)	Stale-frame detection; sensor health; overlay warnings; inhibit auto-track on stale; operator cross-check	Sensor fault injection; frame-freeze tests; field demo
24	Auto-tracking drifts / mis-tracks	Algorithm error; clutter	Operator fires at wrong object	RAC 2 (Serious)	HITL: require explicit operator confirm before FIRE; track confidence gating; easy drop-track control	Tracking evaluation; scenario tests; operator-in-the-loop trials
25	Ballistic solution wrong	Bad zero; wrong ammo profile; wind input	Miss/collateral	RAC 3 (Medium)	Zeroing procedure; ammo profile control; plausibility checks; display assumptions; inhibit if not zeroed	Range zero tests; ballistic regression; configuration tests
26	Ammunition count error	Sensor error; software counter drift	Operator decision error; unexpected empty	RAC 4 (Low)	Dual method (shot count + belt sensor if available); reconcile on reload; warning on low confidence	Counter validation; reload scenario tests
27	Mechanical stop failure	Stop fracture; wear	Over-travel; cable snag; unsafe pointing	RAC 3 (Medium)	Stop strength margin; redundant limit switches; current limiting; periodic inspection	End-stop overload test; inspection; motion tests
28	Cable snag/entanglement with DRONE	Routing error; wear	Control loss; structural damage	RAC 3 (Medium)	Cable management; strain relief; clearance analysis; protective sleeving	Fit-check; vibration test; flight envelope inspection
29	EMI causes control upset	RF emitters on DRONE; payload interference	Unexpected motion; loss of function	RAC 3 (Medium)	EMI shielding; filtering; separation; grounding; software robustness	EMI tests; on-platform integration test
30	Software crash / reboot during operation	Unhandled exception; memory leak	Loss of control; possible unsafe state	RAC 2 (Serious)	Safe-state on reboot; watchdog; state machine recovery; redundancy/backup controller	Stress tests; fault-injection; long-run endurance; reboot recovery demo
31	Clock/time sync error affects command validity	Time drift; GPS time loss	Command acceptance wrong; inhibit fails	RAC 4 (Low)	Monotonic timers; sanity checks; disable time-based gating on invalid time; alarm	Time-drift tests; GPS-loss tests
32	Inadvertent mode change (SAFE/ARM/AUTO)	UI design; accidental click	Unexpected behavior	RAC 2 (Serious)	Guarded mode switch; confirm dialog; logging; audible/visual annunciation	Human factors test; UI validation
33	Maintenance hazard – arming during service	Procedure lapse; service mode missing	Injury to maintainer	RAC 1 (High)	Maintenance lockout/tagout; physical safety pin; service mode disables firing circuits; SOP	Maintenance procedure audit; lockout verification
34	Transport hazard – accidental actuation	Vibration; packed state	Damage/injury	RAC 2 (Serious)	Transport lock; mechanical restraint; SAFE state; covers; shipping procedure	Transport vibration test; pack-out inspection
35	Weapon release/quick disconnect failure	Latch failure; improper assembly	Drop; injury	RAC 2 (Serious)	Positive locking + secondary latch; assembly checklist; visual indicator; torque marks	Drop/load test; assembly verification; inspection
36	Overcurrent in actuator leads to fire	Stall; short; overheating motor	Fire; loss of DRONE	RAC 2 (Serious)	Current limiting; fuses; temperature monitoring; stall detection and shutdown	Stall tests; thermal monitoring validation
37	Shot counter / feedback sensor failure hides firing	Sensor disconnected	Runaway burst undetected	RAC 2 (Serious)	Independent burst limiter; redundant detection (acoustic/accel optional); health monitoring	Sensor disconnect tests; burst limiter demo
38	DRONE flight controller interaction causes oscillations	Coupled dynamics; control mixing	Instability; crash	RAC 2 (Serious)	Integration analysis; limit firing to stable hover; interlock with DRONE attitude bounds; tuning	Flight tests; coupled-model simulation; firing in flight validation

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39	Friendly fire due to wrong target designation	Wrong datalink target; UI mix-up	Injury/death	RAC 1 (High)	HITL positive ID; ROE prompts; target labeling; confirm-before-fire; logs	Operational scenario tests; training; UI validation
40	Loss of video feed while armed	Camera failure; link congestion	Operator blind; potential unsafe	RAC 2 (Serious)	Automatic DISARM if video lost beyond threshold; fallback sensor; alarm; inhibit FIRE without live video	Video loss tests; comm congestion tests; inhibit demo

10. SSPP – Defense Level Narrative

- The System Safety Program integrates hazard identification, analysis, mitigation, verification, and risk acceptance across the lifecycle.
- Safety organization includes Safety Manager, Program Manager, and Independent Safety Authority.
- Safety reviews: SRR, PDR, CDR, TRR, ORR.
- Hazard tracking is maintained in Hazard Log with configuration control.
- Residual risks formally accepted by authority prior to operational release.

11. CONOPS Safety with Fail Safe Logic

- System is human in the loop. Firing requires dual confirmation and valid arm state.
- Fail safe logic includes automatic disarm on communication loss, power loss, or sensor failure.
- Jam detection inhibits firing.
- Mechanical stops limit unsafe pointing.
- Operator must maintain positive identification before engagement.

12. Safety Verification Matrix (Step Procedures)

- Unintended fire prevention test – verify dual confirmation logic
- Communication loss test – verify automatic disarm
- Power interruption test – confirm safe state
- Jam detection test – trigger inhibit verification
- Cyber resilience test – authentication verification
- EMI test – verify immunity
- Structural load test – mount integrity
- Recoil impact flight test – DRONE stability

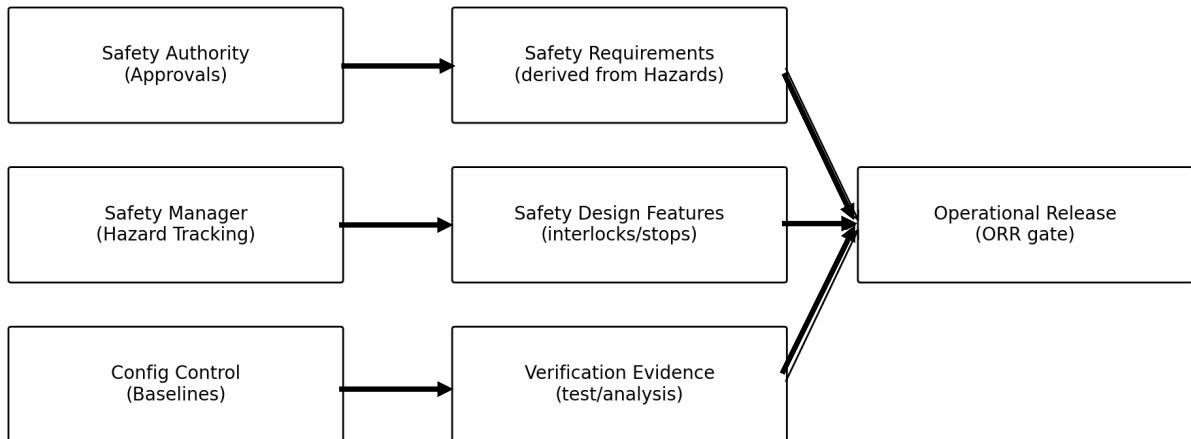
13. Safety Verification Procedures

- Define test configuration and safety observers
- Arm system under controlled range conditions
- Inject failure condition
- Verify automatic safe response
- Record telemetry and logs
- Repeat across operational envelope
- Document results and update hazard log

14. Architecture Safety Diagram

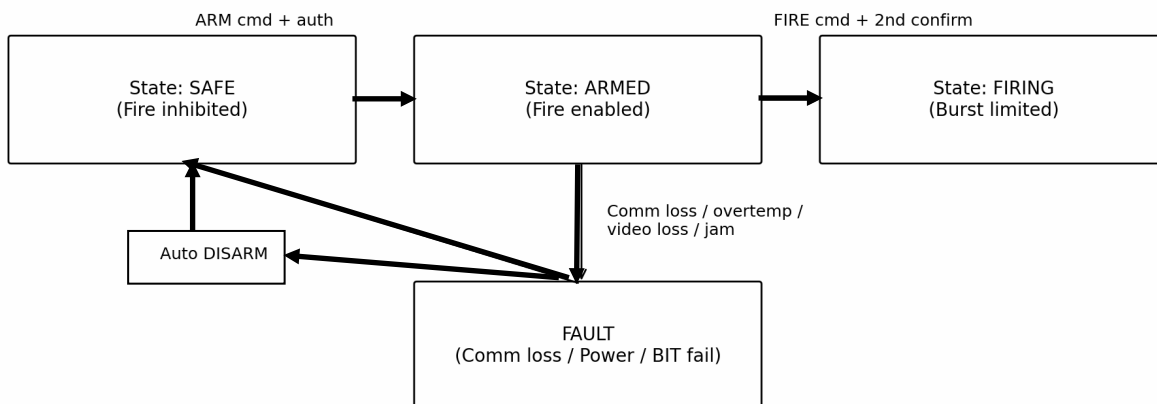
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- Safety governance and evidence flow from hazards to requirements, design, verification, and operational release.



15. Control Logic Fail-Safe Diagram

- High-level safe-state logic for SAFE/ARMED/FIRING with automatic transition to SAFE on faults.



16.Recoil–Flight Coupling Safety Analysis

- Hazard focus: coupled dynamics between recoil impulses, turret stabilization, and DRONE flight controller.
- Key safety requirements (examples): Limit burst length and rate-of-fire such that DRONE attitude rates remain within flight controller bounds.
- Inhibit firing if DRONE attitude/altitude margins are below defined thresholds (e.g., low altitude, high wind).
- Detect stabilization saturation or IMU inconsistency and transition to SAFE state.

17.Applicable Standards & Compliance Matrix

Standard	Scope	Compliance Approach
MIL-STD-882E	System Safety	Hazard Log, RAC, SSPP, SAR integrated

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MIL-STD-810	Environmental qualification	Environmental test matrix and qualification evidence
MIL-STD-461	EMC/EMI	EMI/EMC verification tests and design controls
MIL-STD-1472	Human Factors Engineering	HFE requirements and operator interface safety validation
DoDI 8510.01 / RMF	Cybersecurity	RMF artifact linkage to hazard mitigation
JSSSEH	Software System Safety	Software hazard analysis and verification lifecycle

18. Cyber-Safety Interface (RMF Linkage)

- Cyber hazards including spoofing, replay, unauthorized control and denial of service are mapped to RMF controls and mitigated via encryption, authentication, and monitoring.

19. HFE/HMI Safety Requirements (MIL-STD-1472)

- Operator interface shall support error-resistant design, clear state announcement, alarm prioritization, and dual-confirm firing authorization.

20. EMC/EMI Verification Block (MIL-STD-461)

- System shall undergo conducted and radiated emission and susceptibility testing with mitigation via shielding, filtering, and grounding.

21. Software Safety Process Summary (JSSSEH linkage)

- Software safety integrates hazard-driven requirements, traceability, fault injection testing, independent code review, and verification evidence aligned with JSSSEH.

22. Architecture Partition Safety (ASIL-like Isolation)

- System architecture partitions safety-critical and non-critical domains with isolation barriers, monitored interfaces, and fail-safe transitions.

23. Fire Authorization Chain Safety

- Fire command requires sequential authorization:
Target Detected Target Lock Confirm Target Azimuth Confirm Video in NRT <less 250M/s> Confirm Recoil Management Operation Confirm LRF Operation Confirm Ballistic Calculation Confirm No Ammunition Jam Confirm Drone Stability/Health Confirm Safety Release Confirm arm state Confirm operator safety release Confirm operator authentication Confirm safety interlock enables Confirm Operator trigger actuation Confirm ammunition shot calculation Confirm recoil management operation.